

Case Study: Celestial Object Analyzer

Full-stack Python web app - computer vision + live astronomy catalogs

Source code: github.com/akdemirmert/Celestial-Object-Analyzer

3,739 / 5,705

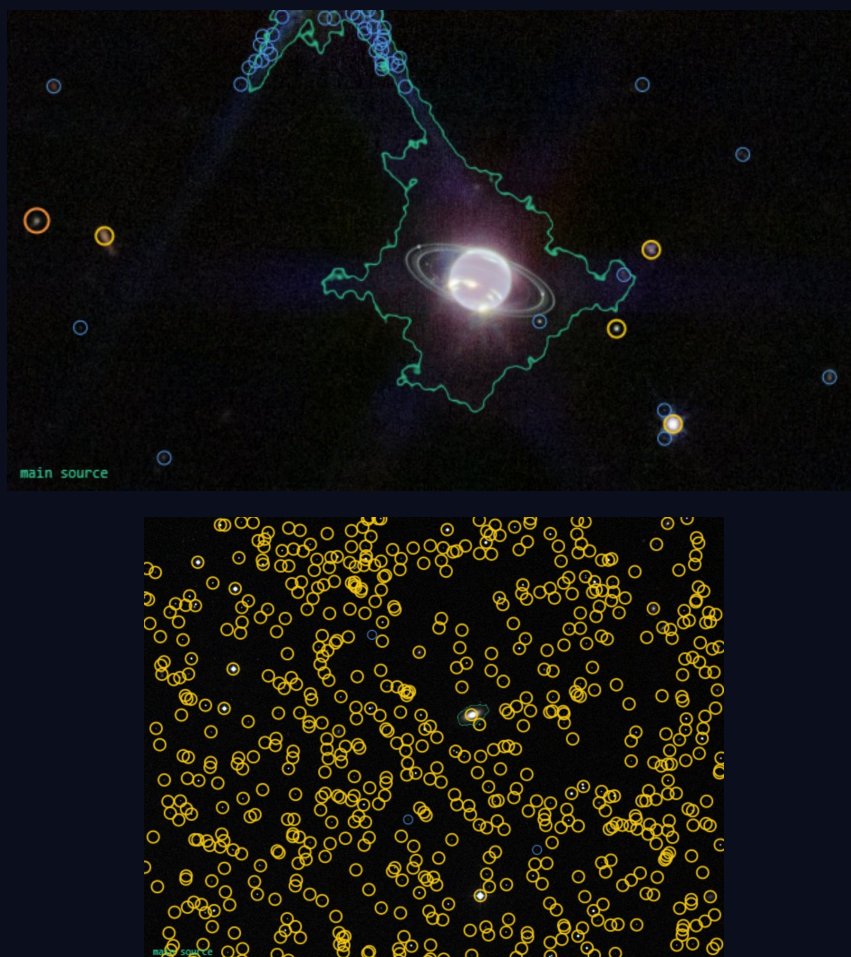
sources named on one JWST frame

0.2"

typical match accuracy (sub-arcsecond)

100% free

runs on zero-cost services & local engine



Top: JWST Neptune frame classified as a ringed planet. Bottom: solved star field with hundreds of catalog-named sources.

What it does

Upload any sky photo - a phone snap or a space-telescope image. The app pins the exact sky position via astrometric plate solving, then names every detected source from live catalogs (SIMBAD, Gaia DR3, Hubble Source Catalog, Pan-STARRS). When identification is impossible it says so honestly, with evidence - it never invents an answer.

Stack

Python - FastAPI - NumPy/SciPy - astropy - PIL - vanilla JavaScript - REST APIs